Problem 1. Which of these sentences are propositions? What are the truth values of those that are propositions?

a) $2 + 3 = 5$
b) Wash your hands
c) Swimming is fun
d) $2x \geq x$
e) 4.3 is an integer

Answer.
a) It is a proposition, and it is true.
b) Not a proposition.
c) Not a proposition, fun is subjective.
d) Not a proposition for all of x.
e) It is a proposition, it is false.

Problem 2. Let p and q be the propositions “Swimming at the shore is allowed” and “Sharks have been spotted near the shore,” respectively. Express each of these compound propositions as an English sentence.

a) $\neg q \land \neg p$
b) $\neg q \to p$
c) $p \leftrightarrow \neg q$

Answer.

a) No sharks have been spotted near the shore, nevertheless, swimming at the shore is not allowed.
b) If no sharks have been spotted near the shore, then swimming at the shore is allowed.
c) Swimming at the shore is allowed if and only if sharks have not been spotted near the shore.

Problem 3. For each of the following sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a) Coffee or tea comes with dinner
b) A password must have at least three digits or be at least eight characters long.
c) The prerequisite for the course is a course in number theory or a course in cryptography.
d) You can pay using U.S. dollars or euros.

Answer.

a) Exclusive or, since restaurants only let you pick either coffee or tea as a drink for a meal.
b) Inclusive or. You can have both three digits and have eight characters in your password.
c) Inclusive or. You can have experience with both number theory and cryptography courses and still take this course.
d) Exclusive or. You cannot mix currencies when paying at a store.

Problem 4. Construct a truth table for each of these compound propositions.

a) $(p \lor \neg q) \to q$
b) $(p \to q) \leftrightarrow (\neg p \lor q)$
c) $(p \land q) \to (p \lor q)$

Answer.

a)

p, q	$\neg q$	$p \lor \neg q$	$(p \lor \neg q) \to q$
$\bot\bot$	$\top$	$\top$	$\bot$
$\bot\top$	$\bot$	$\bot$	$\top$
$\top\bot$	$\top$	$\top$	$\bot$
$\top\top$	$\bot$	$\top$	$\top$

b)

p, q	$p \to q$	$\neg p$	$\neg p \lor q$	$(p \to q) \leftrightarrow (\neg p \lor q)$
$\bot\bot$	$\top$	$\top$	$\top$	$\top$
$\bot\top$	$\top$	$\top$	$\top$	$\top$
$\top\bot$	$\bot$	$\bot$	$\bot$	$\top$
$\top\top$	$\top$	$\bot$	$\top$	$\top$

c)

p, q	$p \land q$	$p \lor q$	$(p \land q) \to (p \lor q)$
$\bot\bot$	$\bot$	$\bot$	$\top$
$\bot\top$	$\bot$	$\top$	$\top$
$\top\bot$	$\bot$	$\top$	$\top$
$\top\top$	$\top$	$\top$	$\top$

Problem 5. State the converse, contrapositive, and inverse of each of the following conditional statements.

a) I will wear a sweater only if it is below freezing.
b) I come to class whenever there is a quiz.
c) If I have a connecting flight, it is necessary for me to fly business class.

Answer.

a) **Converse**: It is below freezing only if I will wear a sweater.
Contrapositive: It is not below freezing only if I won't wear a sweater.
Inverse: I won't wear a sweater only if it isn't below freezing.
b) **Converse**: There is a quiz whenever I come to class.
Contrapositive: There isn't a quiz whenever I don't come to class.
Inverse: I don't come to class whenever there isn't a quiz.
c) **Converse**: If I fly business class, it is necessary for me to have a connecting flight.
Contrapositive: If I don't fly business class, it isn't necessary for me to have a connecting flight.
Inverse: if I don't have a connecting flight, it isn't necessary for me to fly business class.

Problem 6. Determine whether each of the following conditional statements is true or false. Explain your answers.

a) If $1 + 1 = 2$ then $2 + 2 = 5$.
b) If $1 + 1 = 3$ then $2 + 2 = 4$.
c) If $1 + 1 = 2$ then $2 + 2 = 4$.
d) If monkeys can fly then $1 + 1 = 3$.

Answer.

a) False. $p \equiv 1 + 1 = 2 \equiv \top$, $q \equiv 2 + 2 = 5 \equiv \bot$, $\top \to \bot = \bot$
b) True. $p \equiv 1 + 1 = 3 \equiv \bot$, $q \equiv 2 + 2 = 4 \equiv \top$, $\bot \to \top = \top$
c) True. $p \equiv 1 + 1 = 2 \equiv \top$, $q \equiv 2 + 2 = 4 \equiv \top$, $\top \to \top = \top$
d) True. $p \equiv$ monkeys can fly $\equiv \bot$, $q \equiv 1 + 1 = 3 \equiv \bot$, $\bot \to \bot = \top$

Problem 7. Determine whether each of the following biconditional statements is true or false. Explain your answers.

a) $1+1=2$ if and only if $2+2=5$.
b) $1+1=3$ if and only if $2+2=4$.
c) $1+1=2$ if and only if $2+2=4$.
d) Monkeys can fly if and only if $1+1=3$.

Answer.

a) False. $p \equiv 1+1=2 \equiv \top$, $q \equiv 2+2=5 \equiv \bot$, $\top \leftrightarrow \bot = \bot$
b) False. $p \equiv 1+1=3 \equiv \bot$, $q \equiv 2+2=4 \equiv \top$, $\bot \leftrightarrow \top = \bot$
c) True. $p \equiv 1+1=2 \equiv \top$, $q \equiv 2+2=4 \equiv \top$, $\top \leftrightarrow \top = \top$
d) True. $p \equiv \text{monkeys can fly} \equiv \bot$, $q \equiv 1+1=3 \equiv \bot$, $\bot \leftrightarrow \bot = \top$